Technical Report: M/M/1 Queue with Reneging Chucktown Urgent Care

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Abstract

This report extends the Chucktown Urgent Care M/M/1 model to allow impatient patients to leave before being seen. Each waiting patient has an exponentially distributed patience time with mean 15 minutes ($\theta = 4$ per hour). With $\lambda = 2$ and $\mu = 3$, reneging reduces congestion: the mean number in the system drops from 2.000 to 0.618 and the mean wait time drops from 40 minutes to about 4 minutes. However, roughly 29% of patients leave without being treated, as measured by the fraction served $\pi_s \approx 0.709$.

1 Introduction

Staff at Chucktown Urgent Care have observed that patients who wait too long leave and go to a competitor. This report models that behavior by extending the standard M/M/1 queue to include reneging. Each patient who must wait draws an independent exponential patience time. If service has not started by the time that patience expires, the patient leaves without being treated and is recorded as reneged.

2 Model Parameters

All times are in hours. Patients arrive at rate $\lambda = 2$ per hour and are served at rate $\mu = 3$ per hour (20-minute mean service time). The mean patience time is 15 minutes, giving a renege rate of $\theta = 4$ per hour. One doctor is on duty and twenty independent 8-hour shifts are simulated.

3 Theoretical Steady-State Results

3.1 Deriving P_n

This system is a birth-death process. The standard formula for the steady-state probability of state n in terms of P_0 is:

$$P_n = \frac{\lambda_0 \lambda_1 \cdots \lambda_{n-1}}{\mu_1 \mu_2 \cdots \mu_n} P_0 \tag{1}$$

The birth and death rates for this model are:

- Arrivals: $\lambda_n = \lambda$ for all $n \geq 0$, since patients arrive regardless of queue length.
- Departures: $\mu_n = \mu + (n-1)\theta$ for $n \geq 1$. One patient is being served at rate μ and the remaining $n-1$ waiting patients each renege independently at rate θ.

The numerator simplifies immediately since the arrival rate is constant:

$$\lambda_0 \lambda_1 \cdots \lambda_{n-1} = \lambda^n \tag{2}$$

Substituting the death rates for states 1 through n gives the denominator:

$$\mu_1 \mu_2 \cdots \mu_n = \mu\,(\mu + \theta)\,(\mu + 2\theta) \cdots (\mu + (n-1)\theta) \tag{3}$$

Putting these together:

$$P_n = \frac{\lambda^n}{\mu\,(\mu+\theta)\,(\mu+2\theta)\cdots(\mu+(n-1)\theta)}\,P_0, \quad n \geq 1 \tag{4}$$

Written compactly using product notation:

$$\boxed{P_n = \frac{\lambda^n}{\prod_{k=0}^{n-1}(\mu+k\theta)}\,P_0, \quad n \geq 1} \tag{5}$$

3.2 Computing P_0 and π_s

$$P_0 = \frac{1}{F(1;\,\mu/\theta;\,\lambda/\theta)} \tag{6}$$

where F is the hypergeometric function, computed in MATLAB as `1/hypergeom([1],[mu/theta],lambda/theta)`. The fraction of patients eventually served rather than reneging is:

$$\pi_s = \frac{\mu(1-P_0)}{\lambda} \tag{7}$$

3.3 Theoretical Values

Table 1 compares the theoretical state probabilities with and without reneging.

Table 1: Theoretical State Probabilities: Reneging vs. Baseline

State n	**No Reneging**	**With Reneging**	**Rel. % Diff**
0	0.3333	0.5272	+58.2%
1	0.2222	0.3514	+58.1%
2	0.1481	0.1004	−32.2%
3	0.0988	0.0183	−81.5%
4	0.0658	0.0024	−96.4%
5	0.0439	0.00026	−99.4%

P_0 and P_1 rise substantially because patients leave rather than letting the queue grow. The probability of 3 or more patients drops by over 80%. Table 2 gives the theoretical performance measures.

Table 2: Theoretical Performance Measures with Reneging ($\lambda = 2$, $\mu = 3$, $\theta = 4$)

Metric	**Theory Value**
L (mean in system)	0.6182
L_q (mean waiting)	0.1454
W (mean time in system)	0.3091 hr
W_q (mean wait time)	0.0727 hr
π_s (fraction served)	0.7092

4 Simulation Results

The simulation generates histograms of time in system, time in queue, time in service, customers served per shift, and customers who reneged per shift. The distribution of customers in the system is plotted

Table 3: Reneging Performance: Theory vs. Simulation ($\lambda = 2$, $\mu = 3$, $\theta = 4$, $s = 1$)

Metric	**Theory**	**Simulation**	**Rel. % Diff**
L	0.6182	0.6052	−2.10%
L_q	0.1454	0.1354	−6.83%
W (hr)	0.3091	0.3585	+15.98%
W_q (hr)	0.0727	0.0436	−39.97%
π_s	0.7092	0.7235	+2.01%

with theoretical P_0 through P_5 overlaid as scatter points. Table 3 compares theory against the simulation averages.

L, L_q, and π_s agree with theory to within 7%, which is a much smaller difference versus the 15 to 37% difference in the baseline. W is overestimated by 16% and W_q is underestimated by 40%. These errors suggest the simulation's estimate of served and reneging patients shifts the wait-time distribution in a way that affects the time measures more than the count-based ones. With only 20 samples this level of variance in the time measures is expected.

5 Reneging vs. No Reneging

Table 4 summarizes how each metric changes when reneging is introduced.

Table 4: Baseline vs. Reneging Comparison (Theory)

Metric	**No Reneging**	**With Reneging**	**Rel. % Diff**
P_0	0.3333	0.5272	+58.2%
L	2.0000	0.6182	−69.1%
L_q	1.3333	0.1454	−89.1%
W (hr)	1.0000	0.3091	−69.1%
W_q (hr)	0.6667	0.0727	−89.1%
π_s	1.0000	0.7092	−29.1%

With reneging the system appears far less congested. L drops by 69%, L_q by 89%, and wait times decrease dramatically. They occur because roughly 1 in 3 patients gives up and leaves rather than because the clinic has more capacity. A low W_q combined with a low π_s signals that the queue is short because patients are leaving, not because service is adequate. Inspecting π_s alongside the traditional metrics is important to see how reneging affects the urgent care queue.